



STAR RATING
FOR SCHOOLS

Star Rating for Schools

Coding guide



About Star Rating for Schools

Approximately 500 children are killed in preventable traffic crashes around the world every day. Sometimes they are killed just yards from the school entrance. Every traffic death and every injury violate a child's right to an education.

Star Rating for Schools is an evidence-based tool for **measuring, managing and communicating** the risk children are exposed to on a journey to school. It supports quick interventions that save lives and prevent serious injuries from day one.

The United Nations has included road safety in its Sustainable Development Goal target 3.6, which is to halve road deaths and serious injuries by 2030. SR4S is key to meeting these global targets for child and adolescent pedestrians.

Star Rating for Schools was awarded a prestigious [Prince Michael International Road Safety Award](#) in December 2018, recognising the application's huge potential to reduce risk and save the lives of pedestrians worldwide.

Star Rating for Schools harnesses the power of the iRAP Star Ratings to measure the risk children are exposed to on their journey to school. The iRAP Star Ratings are an internationally recognised measure of risk on roads, with the least safe roads rated as one star and the safest as five stars. Star Rating for Schools utilises the pedestrian component of Star Ratings to provide a measure of the contribution of road design to the risk for each pedestrian.

Star Rating for Schools has only been made possible through the generous support of:

Founding sponsor:



Major donor:



Global programme partner:



For more information

To find out more about the Star Rating for Schools visit www.starratingforschool.org

For general enquiries, contact iRAP at icanhelp@irap.org

Star Rating for Schools Coding Guide Version 1.7 (English)

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1 Introduction

This coding guide provides a guide for all road features that are collected as part of Star Rating for Schools (SR4S). Further information on iRAP Star Rating methodology and access to training courses can be found at www.irap.org/training.

2 New location

2.1 Road name

The name of the primary road where the assessment is being conducted.

2.2 Section

The section name/number is used to differentiate between different locations along the same road, for instance. It is recommended that the section name/number be meaningful to the end user. For example, section names could describe a location as 'in front of school'.

2.3 Assessor comment

If there are key context comments in relation to the location, these should be recorded. For example: 'school main entrance', 'shortcut used by children', 'bus stop', etc.

2.4 Right and left side of the road

The terms 'left side' and 'right side' are used throughout the manual. To define each side for the codification, it is suggested to consider the direction of traffic (imagine yourself as a driver and use this direction as the reference). If for some reason you have coded a specific location using a different reference, please add this as a note in the comments section and provide images registering this coding perspective.

2.5 Image

Add images of the location that are sufficient to allow the data reviewer to check all the data collected for accuracy. Pictures are also important for keeping records of the data collection process and decisions made by the assessor in the field. For example, right and left side of the road, special conditions, etc. These will also be helpful in capturing the baseline situation and will allow a comparison to be made after road upgrades.



3 Road environment

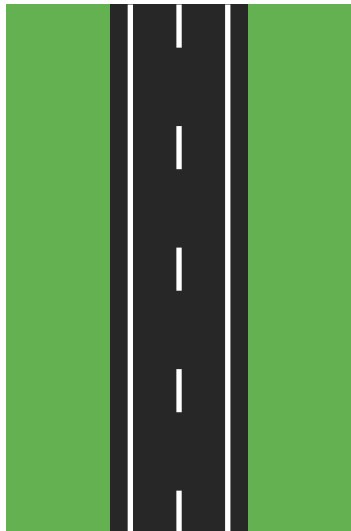
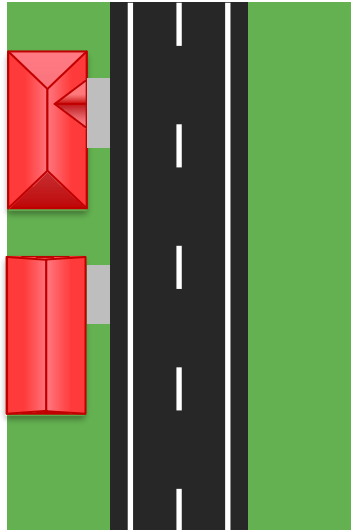
3.1 Land use

Record the type of roadside development that is observed on right and left side of the road. This is used to provide an indication of the number of pedestrians likely to be in the area.

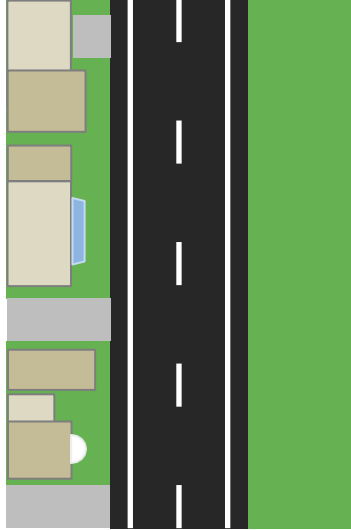
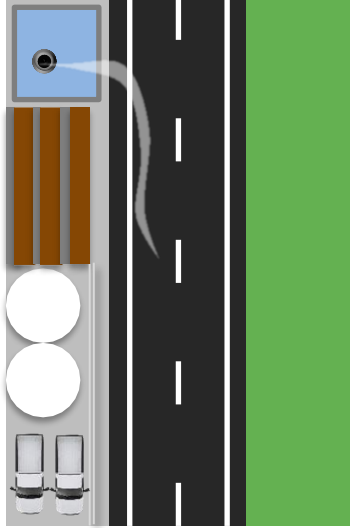
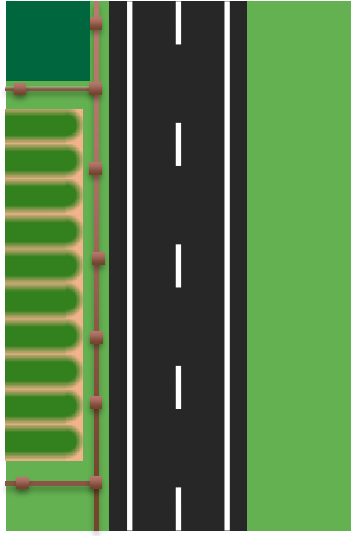
Notes:

Any areas of obvious potential high-intensity pedestrian activity should be recorded as 'commercial' even if commercial activity is not immediately obvious.

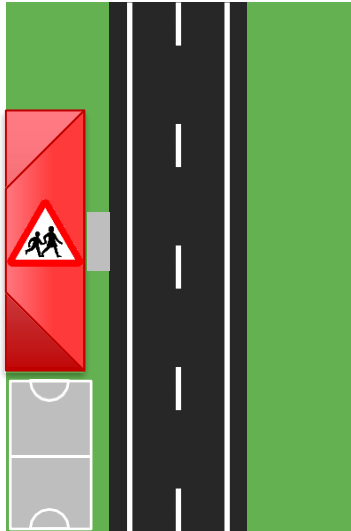
Coding options

	Undeveloped	
	Open land without shops, housing, industry or agriculture. This should be recorded where the road is separated from its environment with no possibility of the presence of pedestrians, even in an urban area, for example, a bridge.	
	Residential	
	Residential housing is present.	



	Commercial	
	Shops or other commercial activity, parks and recreation spaces (such as sporting fields) or areas where high intensity pedestrian activity is evident.	
	Industrial	
	Industrial units, factories and manufacturing sites.	
	Farming	
	Land used for farming or agricultural activities.	



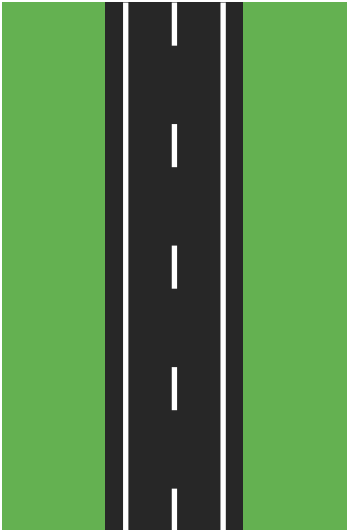
	School	
Schools, colleges and universities.		

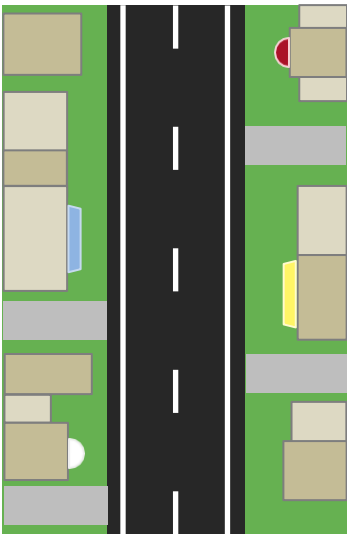


3.2 Area type

Record the level of roadside development through which the road is passing.

Coding options

	Rural / open area	
Area outside of an urban area or rural town or village.		

	Urban / rural town or village	
Development, commercial activity, public services or group of houses that impact the road and may influence the presence of pedestrians.		



3.3 Vehicle parking

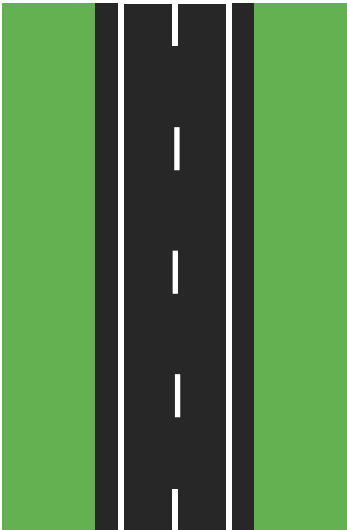
Record the extent to which there is vehicle parking along the edge of the road.

Notes:

This attribute should include parked vehicles regardless of whether they are authorised or not, as well as on-street parking spaces, bus stops and general encroachment on the road.

Vehicle parking should be based on activities within 2m of the outside edge of the driveable lane.

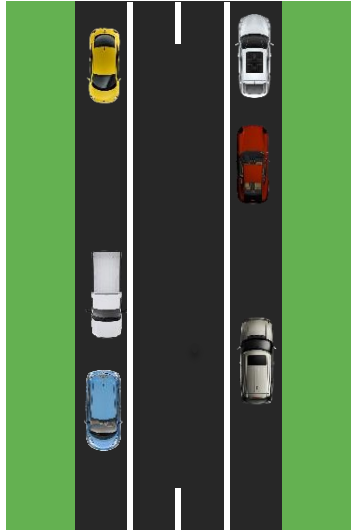
Coding options

	None	
No parking on either side of the road.		

	One side	
Parking on one side of the road.		



	Two sides
Parking on both sides of the road.	





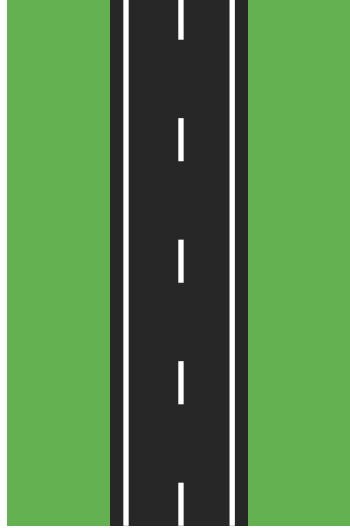
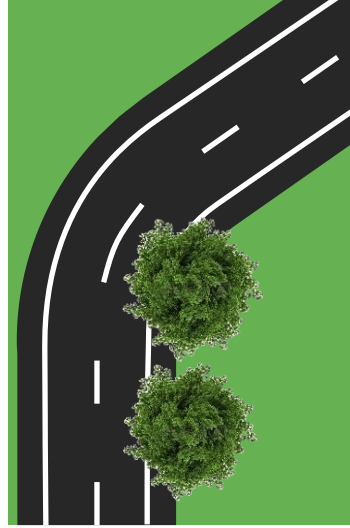
3.4 Sight distance

Record the ability of the driver to see the pedestrians or bicyclists either on the roadway, or about to cross the road ahead, or other vehicles at an intersection.

Notes:

Poor visibility may result from road curves, inclines or declines (e.g. hills or dips in the road) as well as physical obstructions such as roadside objects and vegetation.

Coding options

	Adequate	
	No visibility restrictions along the road.	
	Poor	
	Major/ limited visibility restriction(s) along the road.	



4 Road type

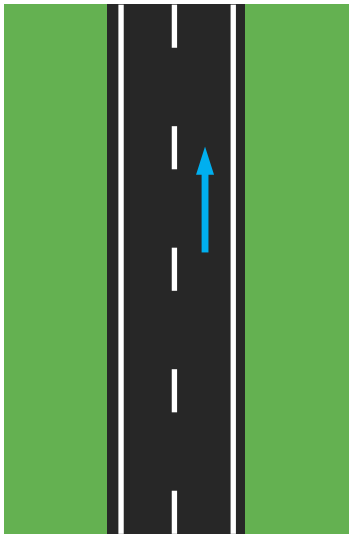
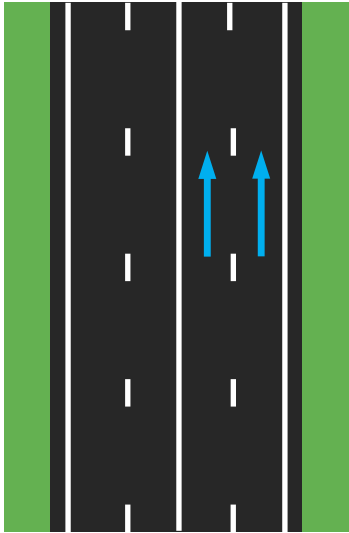
4.1 Number of lanes

Record the number of lanes in each direction of travel (i.e. the number of lanes on one side of the road).

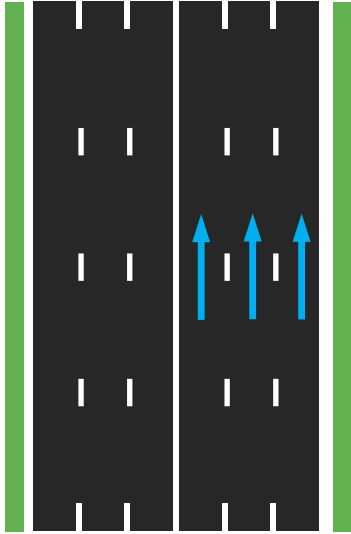
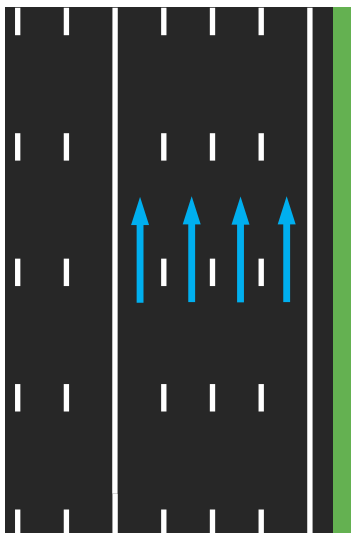
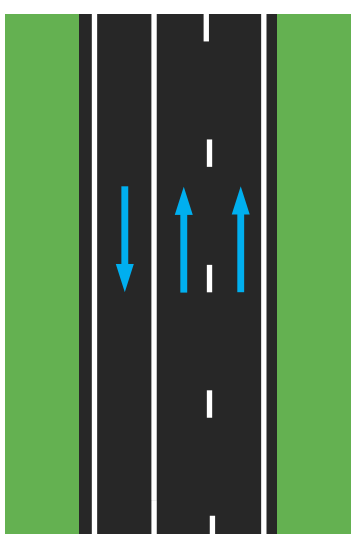
Notes:

Only lanes that serve through traffic should be considered.

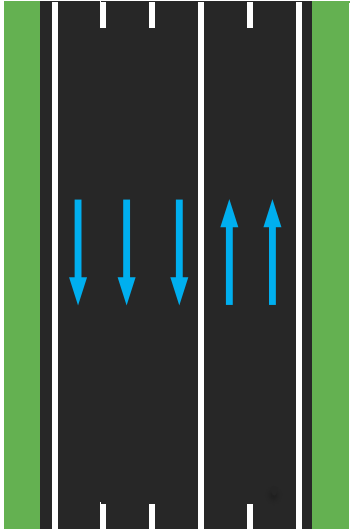
Coding options

1	One	
	One lane in direction of travel.	
2	Two	
	Two lanes in direction of travel.	



3	Three	
	Three lanes in direction of travel.	
4	Four or more	
	Four or more lanes in direction of travel.	
1&2	One and two	
	Undivided carriageway with two lanes in one direction and one in the other direction.	



2&3	Two and three	
	Undivided carriageway with three lanes in one direction and two in the other direction.	



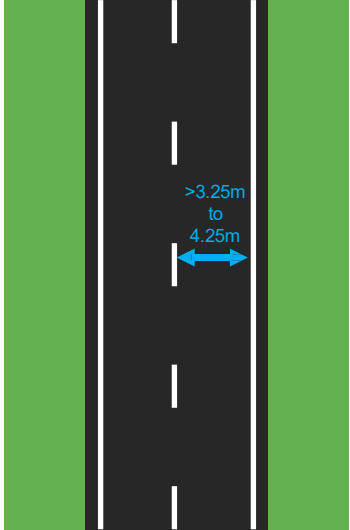
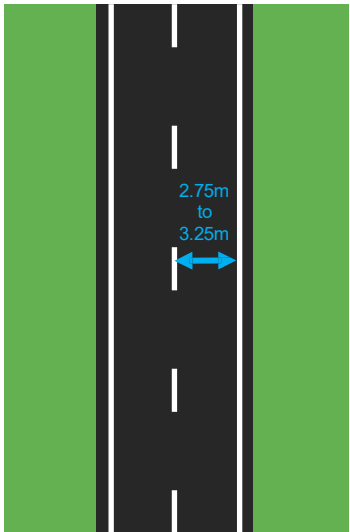
4.2 Lane width

Record the width of each lane.

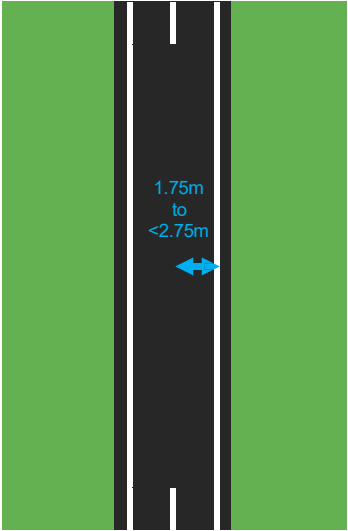
Notes:

If no lane markings are present, the total road width should be divided by the number of traffic streams.

Coding options

WIDE	Wide	
Lane width is greater than 4.25m		
MEDIUM	Medium	
Lane width is 2.75m to 3.25m		



NARROW	Narrow	
	Lane width is less than 2.75m Rule of thumb - A bus or truck would struggle to fit in a narrow lane.	



4.3 Shoulder rumble

Record if the markings on the edge of the road have elements that rumble when vehicles go over them. These will make the vehicle shake and warn drivers that they are leaving the road.

Notes:

Shoulder rumble strips may be constructed by cutting grooves into the pavement surface or by adding raised ribs to the road marking.

Coding options

☐	Not present	
No shoulder rumble strip present.		
☒	Present	
Shoulder rumble strip present.		



4.4 Road surface condition

Record the condition of the road surface.

Good road condition is where there is a level, even running surface that is free from major surface defects that may adversely affect the vehicle path.

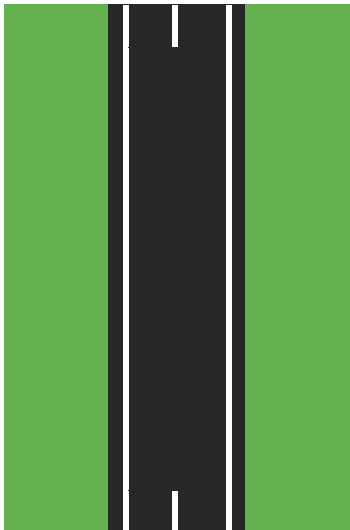
Notes:

Issues that should be considered include anything that would cause an impact on vehicle control, for example:

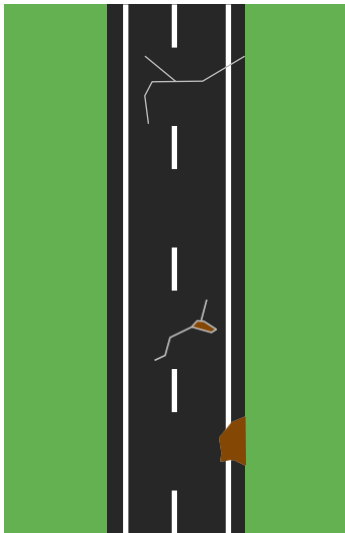
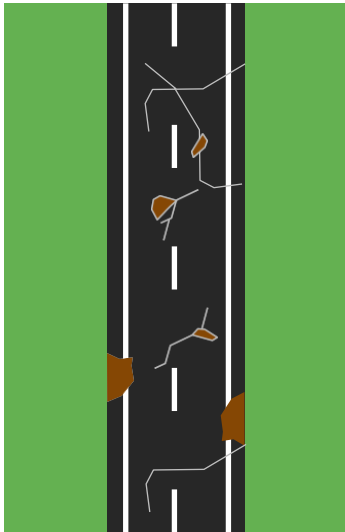
- Deformation and cracking – an uneven surface that can lead to an uncomfortable and/or unsafe journey
- Pot-holes – any holes in the road surface due to loss of material, sufficiently wide or deep enough to cause a severe jolt or loss of control to the vehicle, or
- Edge defects – any problems with the edge of the road which encroach on the driving lane.

The features of a poor or medium-condition road need only to be present at some point along the length, not over its entire length. Typically, a poor category will have serious defects for over 5 meters or more.

Coding options

GOOD	Good	
Road has very few or no issues. There is no potential impact on vehicle control or path.		



MEDIUM	Medium	
Road has a number of minor issues which may result in occasional impact on vehicle control or path.		
POOR	Poor	
Road has serious issues which are likely to result in frequent or unpredictable impact on vehicle control.		



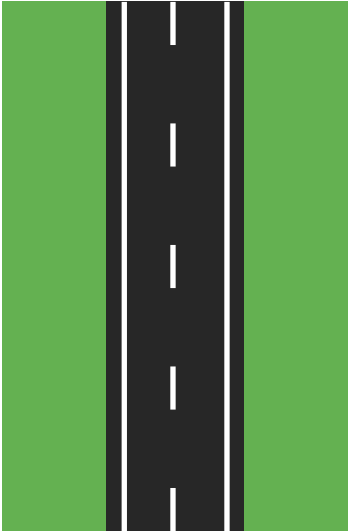
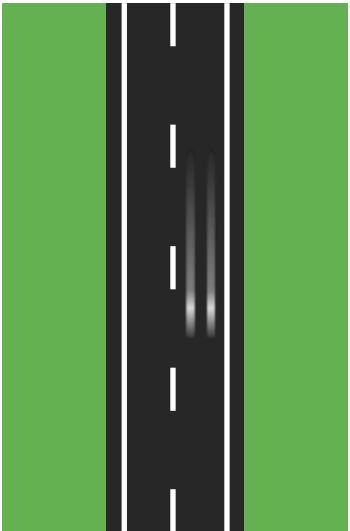
4.5 Grip

Record the level of general grip a vehicle is likely to have on the road.

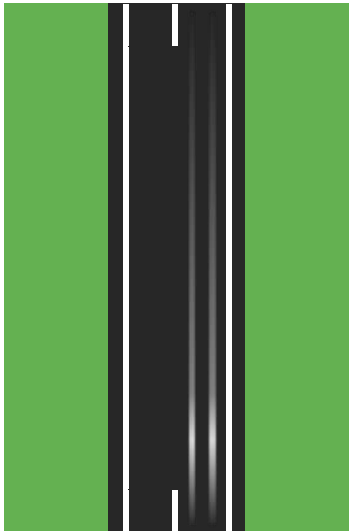
Notes:

Observe issues such as loose gravel as well as sections of the road surface which are smooth, shiny or slippery.

Coding options

GOOD	Good	
The road surface is expected to have adequate skid resistance performance. There are no visible smooth and shiny sections on the preferred vehicle path.		
MEDIUM	Medium	
The road surface has a medium grip surface. For example: • The road surface is paved and looks smooth and shiny for up to 20% of the preferred vehicle path. • Loose gravel or other material is present for up to 20%.		



POOR	Poor	
The road surface has a low grip surface. For example: • The road surface is paved and looks smooth and shiny for more than 20% of the preferred vehicle path. • Loose gravel or other material is present for more than 20%.		



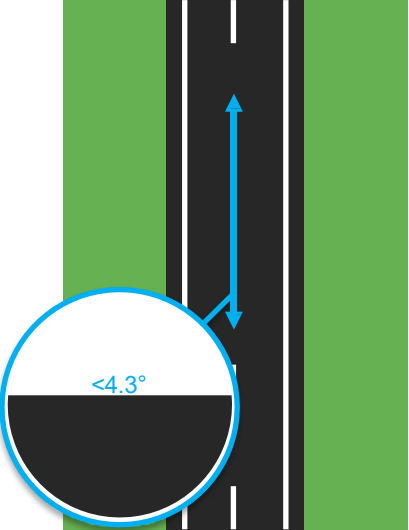
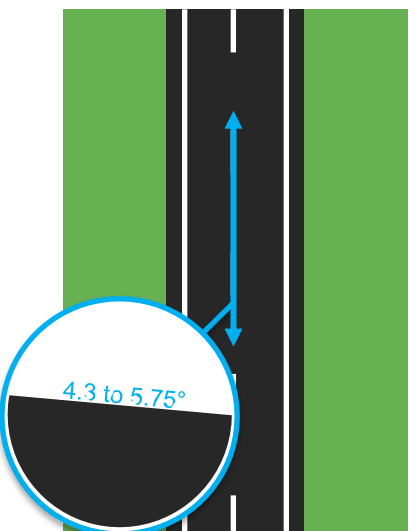
4.6 Grade

Record the slope / gradient of the road along its length.

Notes:

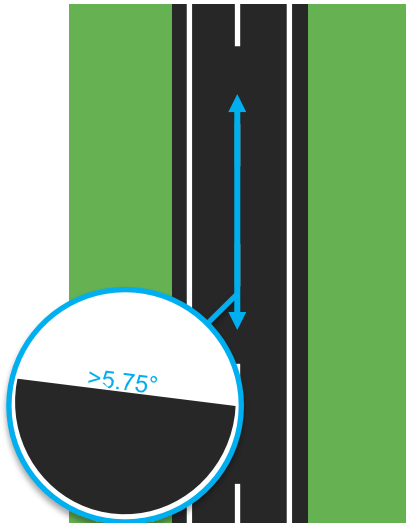
Grade refers to both upward and downward slopes.

Coding options

	≥ 0 to $< 7.5\%$	
	Flat or a rise of up to 7.5m over 100m length, or an angle of up to 4.3 degrees.	
	≥ 7.5 to $< 10\%$	
	A rise of 7.5m to < 10m over 100m length, or an angle of 4.3 to 5.75 degrees.	



	≥10%
A rise of more than 10m over 100m length, or an angle of greater than 5.75 degrees.	





4.7 Carriageway type

Divided (dual) carriageways should be assessed in each direction separately, but undivided (single) carriageway roads are recorded in one direction only, even if the traffic is two-way. What is considered divided and undivided depends on the median type and its length:

- Divided carriageways are those that physically separate opposing traffic flows by either a barrier or a wide physical median consistently and for a distance of 400m or more.
- An undivided carriageway has no physical separation between opposing traffic flows, or physically separates traffic for a section of less than 400m.

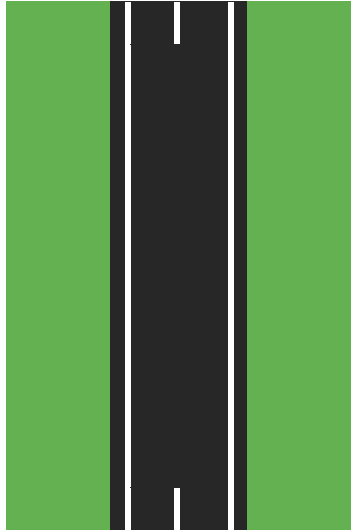
Notes:

One-way roads must be coded as undivided and the 'Median type' set to one-way. Service roads must be coded separately to the main carriageways. Code service roads the same way as standard roads.

Coding options

Divided A	North/East	
Carriageway A assessed.		
Divided B	South/West	
Carriageway B assessed.		



 Undivided	Not Applicable	
Undivided or simple carriageway. There is no consistent central median dividing the two traffic directions. This type of carriageway also includes one-way roads.		



5 Road features

5.1 Median (Middle of road)

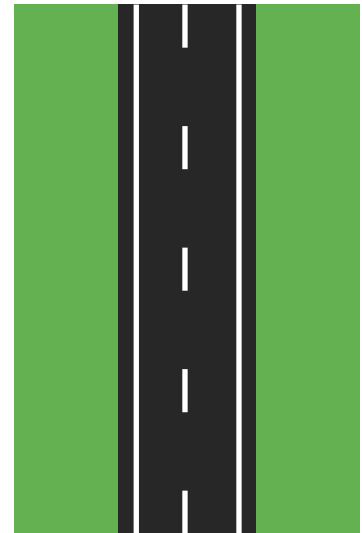
Record the feature that separates the two opposing traffic flows.

Coding options



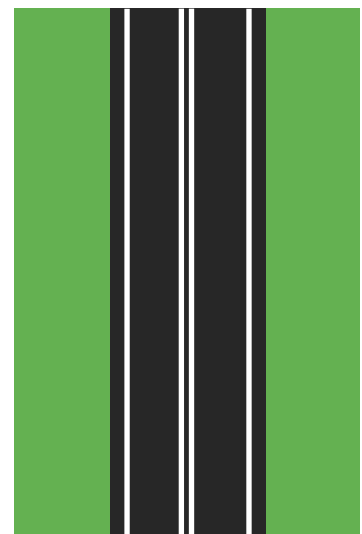
Centreline only

Single or double central line or marked central strip <0.3m wide.
Centre line can be either dashed or solid, and of any colour (according to local regulations).

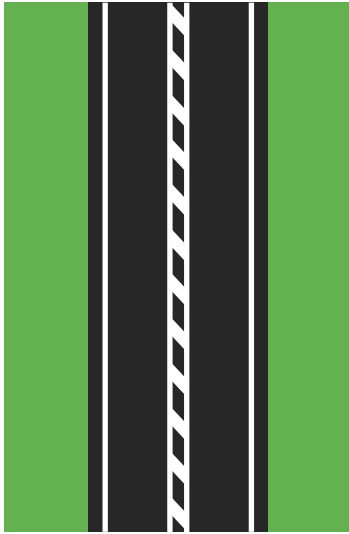
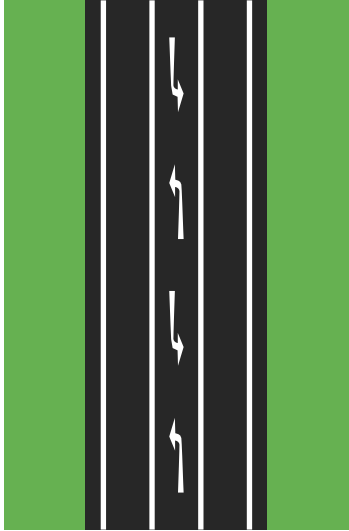
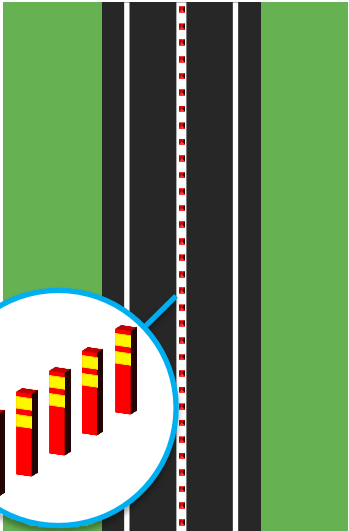


Double centreline

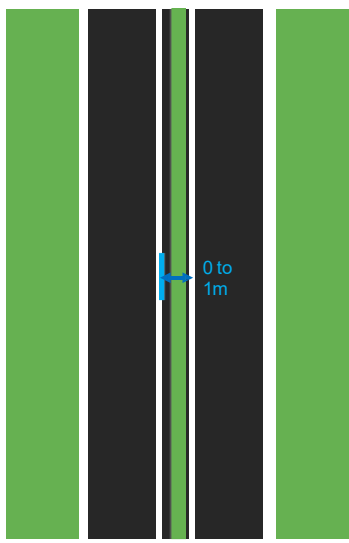
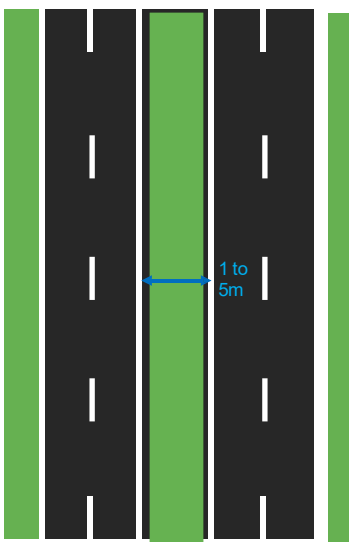
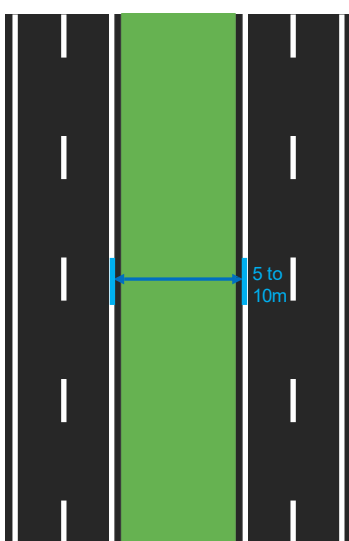
Single or double centre line or marked central strip 0.3m to 1m wide.
Central line can be either dashed or solid, and of any colour (according to local regulations).





	Continuous wide median markings (>60cm)	
	Marked central strip >0.6m wide. Examples of marking are: • Ladder hatching • Stripes • Coloured paving	
	Turn lane	
	Continuous central lane designated as a turning lane.	
	Flexible posts	
	Flexible posts or bollards, but not a continuous physical divider.	



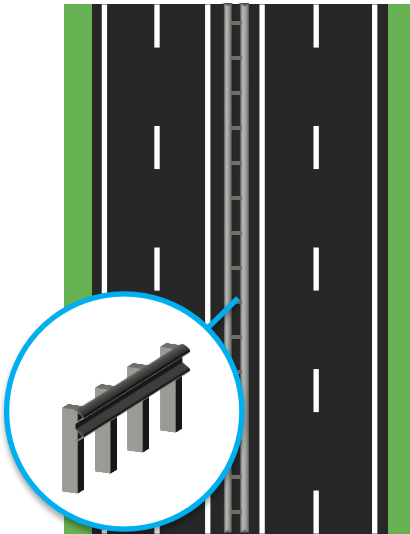
	Separated 0m to <1m	
Divided road with defined median of different material of width <1m.		
	Separated 1m to <5m	
Divided road with defined median of different material of width 1m to <5m.		
	Separated 5m to <10m	
Divided road with defined median of different material of width 5m to <10m.		

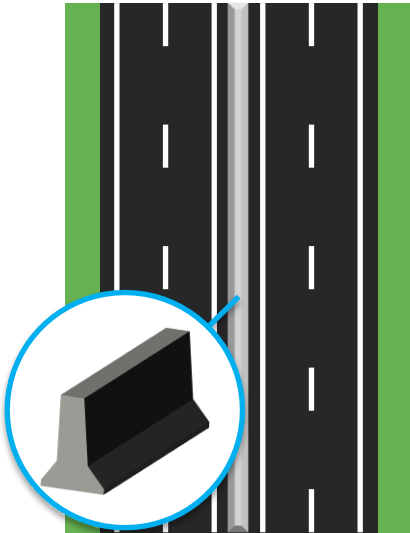


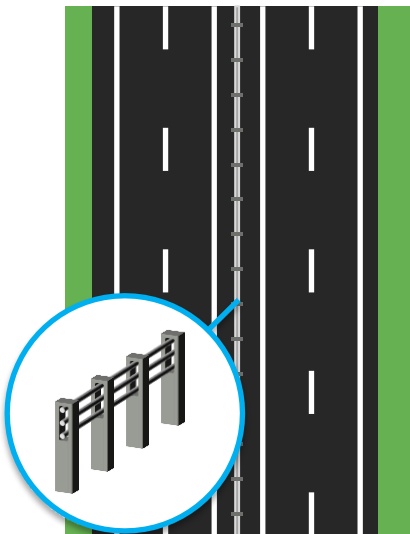
Separated 10m to <20m Divided road with defined median of different material of width 10 to <20m.	
Separated 20+m Divided road with defined median of different material of width 20m or greater.	



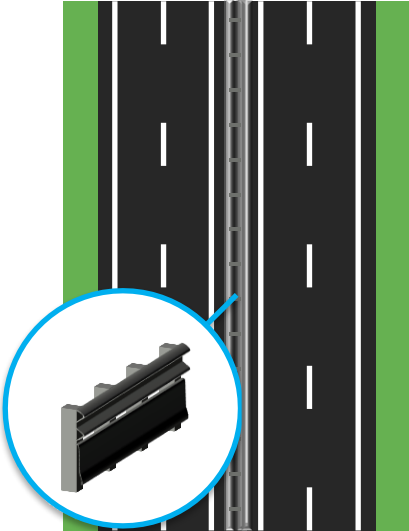
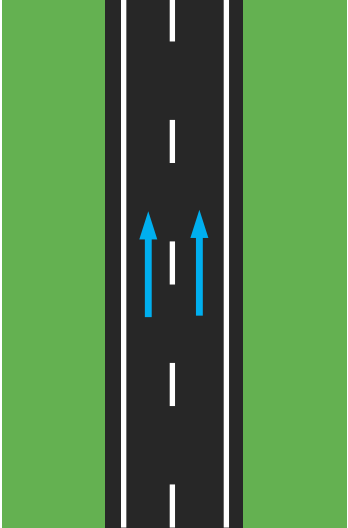
	Metal Barrier
Divided road with safety barrier made of metal. If the barrier is visibly defective, ignore the barrier.	



	Concrete Barrier	
Divided road with safety barrier constructed from concrete. If the barrier is visibly defective, ignore the barrier.		

	Wire Barrier	
Divided road with safety barrier comprised of posts with tensioned wires between them. If the barrier is visibly defective, ignore the barrier.		



	Motorcycle Barrier	
	Divided road with safety barrier with some form of protection for the supporting post or legs. If the barrier is visibly defective, ignore the barrier.	
	One way	
	Vehicles travel in a single direction with no opposing flow.	



5.2 Lines and signs

Record the presence / condition of the road signs, lines and markings.

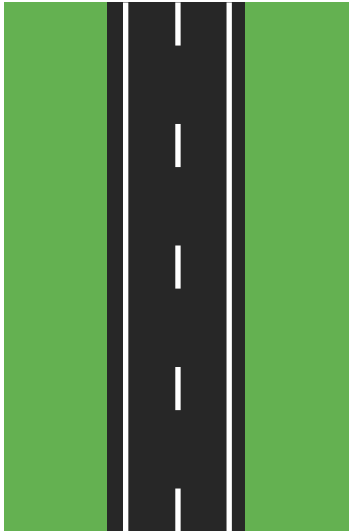
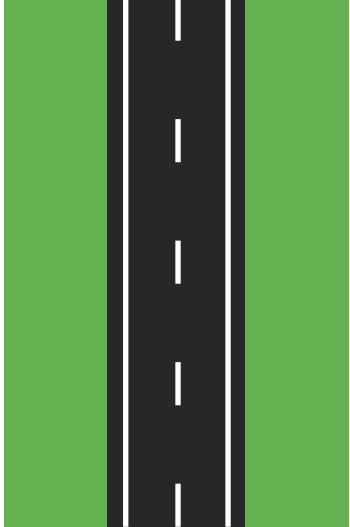
Notes:

Examples of delineation are:

- Centre lines and lane markers
- Edge lines
- Guideposts / delineators, road studs and hazard markers
- Signage

Warning signs must be present to warn drivers of any sudden change in the road conditions like lane narrowing.

Coding options

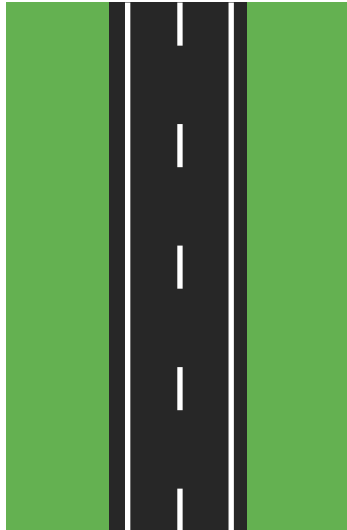
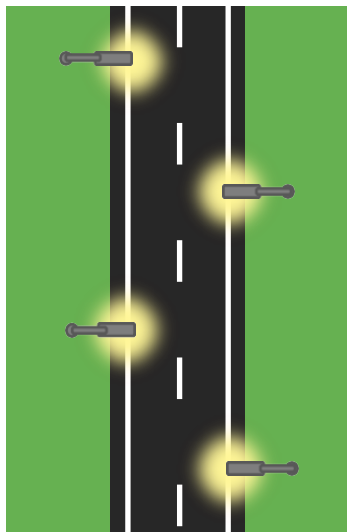
	Adequate	
	Warning signs of severe hazards, and centre and edge markings are generally present and visible.	
	Poor	
	Signing of hazards, or centre and edge markings are absent or in poor condition over long lengths.	



5.3 Street lighting

Record the presence of street lighting that is sufficient for illuminating pedestrians and bicyclists.

Coding options

	Not Present	
Street lighting is not present or insufficient to illuminate pedestrians and bicyclists.		
	Present	
Street lighting is present to a degree sufficient to illuminate pedestrians and bicyclists.		



6 School zone

6.1 School warning

Record the presence of a school zone.

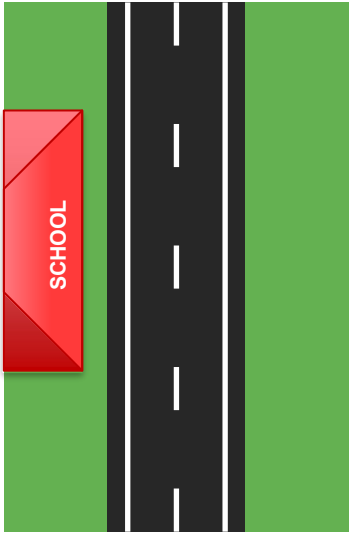
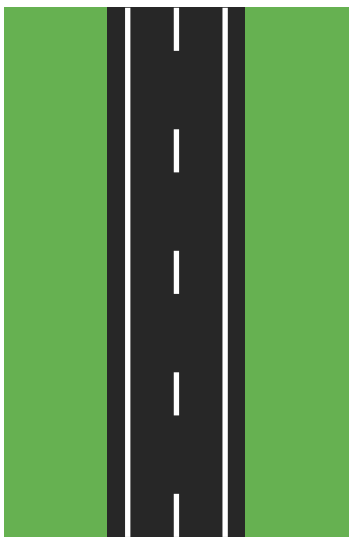
Notes:

School zones are likely to have appropriate road signs and markings to make motorists aware of the presence of vulnerable road users such as young pedestrians and cyclists. Parking restrictions may also apply in school zones. School zones are likely to have reduced speed limits for certain times of the day.

Coding options

Flashing Beacon A school zone is present and incorporates flashing beacons to give emphasis to the warning signs and appropriate speed limit.	
Signs / markings A school zone is present with appropriate warning signs. Road markings and speed limit signs may also be present.	



	No school zone warning	
	School is present but no school zone warning signs or markings are evident.	
	Not applicable	
	No school at the location.	



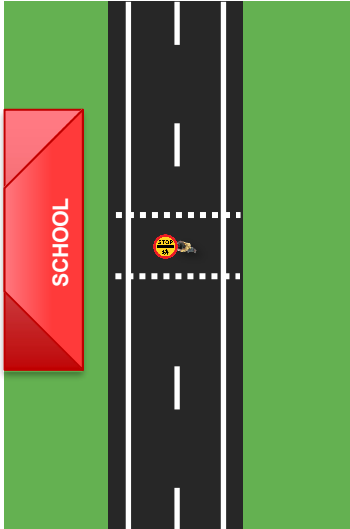
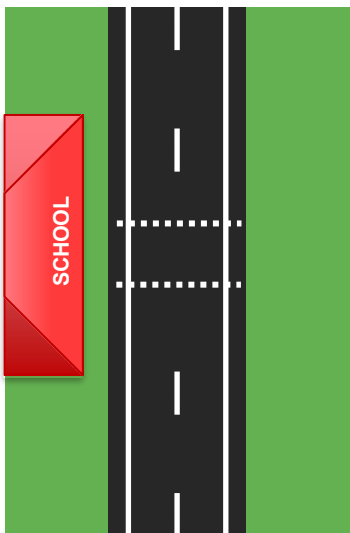
6.2 School crossing supervisor

Record the presence of a crossing supervisor, warden or lollipop lady/man. The crossing supervisor will assist children in safely crossing roads on their way to and from school.

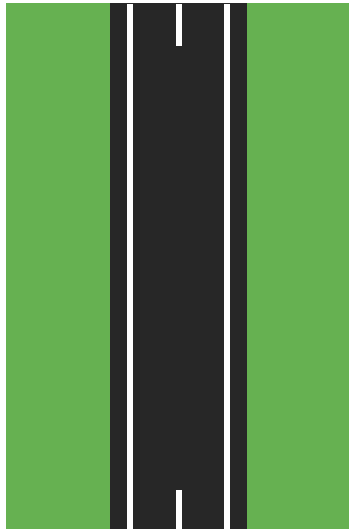
Notes:

School crossing supervisors often operate during the school community's peak demand period both in the morning and in the afternoon. Also referred to as a school crossing patrol.

Coding options

	Supervisor	
The pedestrian crossing has a crossing supervisor present during school hours. The crossing supervisor may be present throughout the school day or during periods of high pedestrian activity only, for example, at school start and finish times.		
	No Supervisor	
No crossing supervisor is present at the pedestrian crossing facility or only present on an irregular basis.		



	Not applicable	
No school at the location.		



7 Sidewalks

7.1 Sidewalks

Record the presence of a pedestrian path on the right and left sides of the roadway.

Notes:

A good quality sidewalk is a purpose-built facility for pedestrians that has an all-weather surface that provides a reliable surface for pedestrians to walk on.

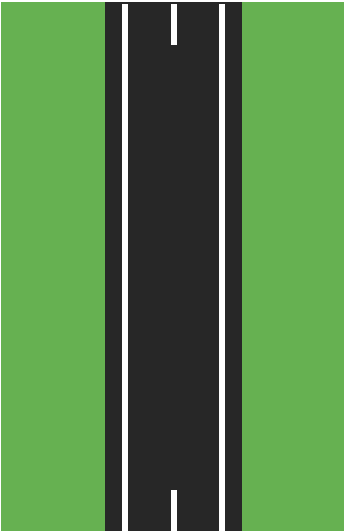
Sidewalk separation is based on its distance from the outer edge of the road (e.g. the curb or the edge of the tarmac).

Curbs are not a physical barrier. While discouraging traffic from passing over them, curbs do little to prevent road traffic from entering the sidewalk, unless specifically designed to do so..

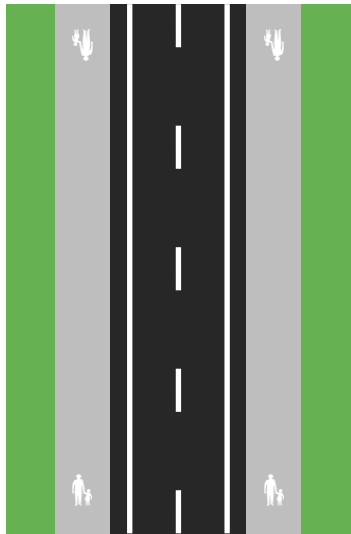
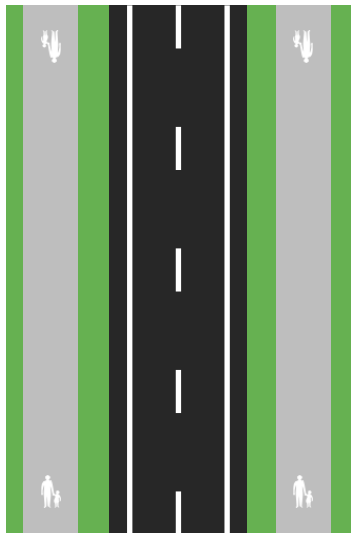
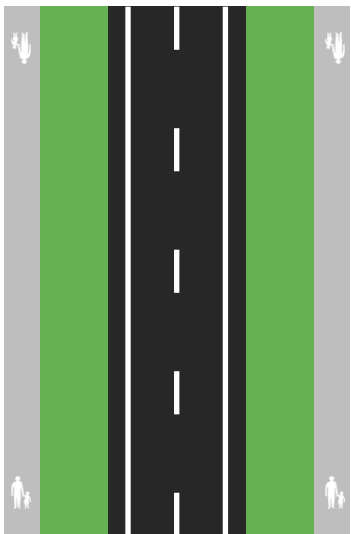
If the sidewalk is obstructed or overcrowded (e.g. by roadside traders or shop front activities) so that pedestrians need to walk in the roadway, then code as if sidewalk is not present.

Non-physical separation may include grass verges.

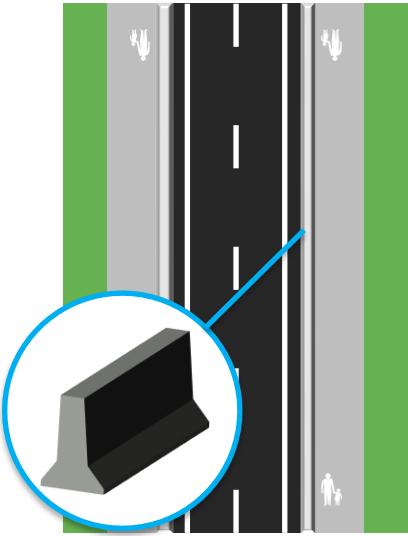
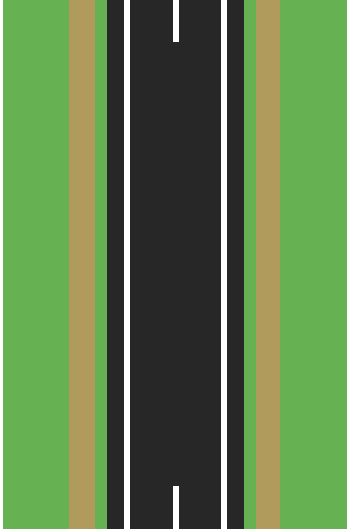
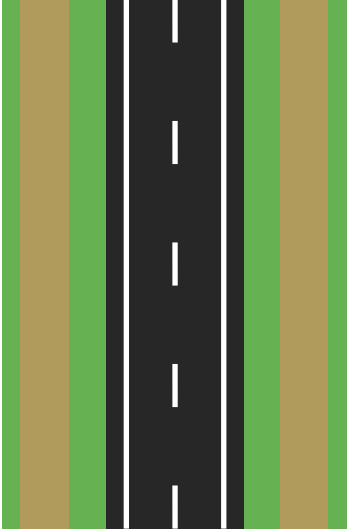
Coding options

	None	
No discernible sidewalk is provided.		

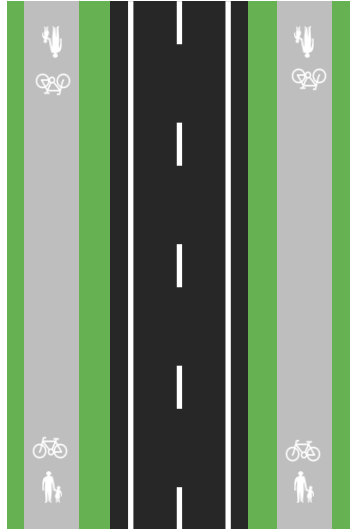


	Sidewalk 0m to <1m Away	
	Good quality sidewalk is adjacent to the road or separated from road by less than 1m.	
	Sidewalk 1m to 3m Away	
	Sidewalk ≥ 3m Away	



	Sidewalk Behind Barrier	
	Sidewalk separated from road by a physical barrier sufficient to restrain a vehicle from entering the pedestrian facility at the posted speed limit.	
	Poor Quality Sidewalk	
	A path which is partially obstructed, has an uneven surface, is vulnerable to weather conditions, has limited access (e.g. very narrow or no ramp), or is less than 1m from the road edge.	
	Medium Quality Sidewalk	
	A path which provides satisfactory access for most pedestrians in fine weather conditions, is unobstructed and is more than 1m from the road edge.	



	Shared Path	
	A path which is shared with bicycles or other types of light mobility.	

7.2 Road edge

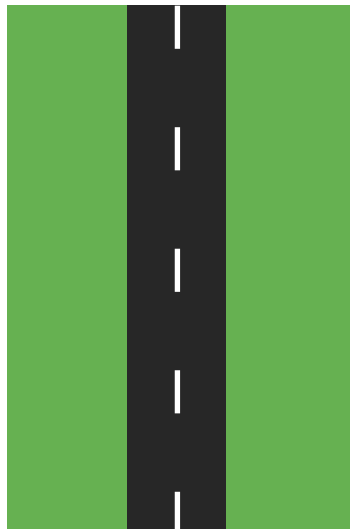
Record the width of the paved road shoulder for both sides of the road, i.e. the distance between the edge line of the outer traffic lane and the curb or edge of the tarmac.

Notes:

If the tarmac begins to break up on a shoulder, the width should be measured up to where the edge break occurs.

If a road has no markings for a shoulder (i.e. no edge line), then record it as 'none' as a vehicle is considered to be able to travel to the edge of the sealed surface.

Coding options

	None	
	There is no edge line present.	



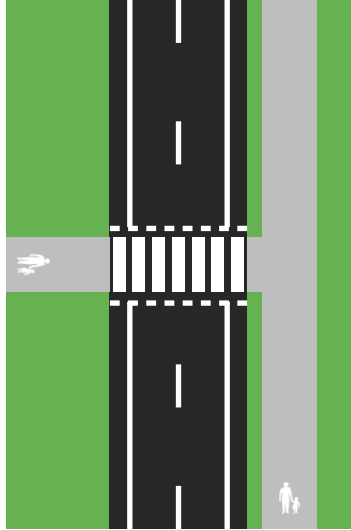
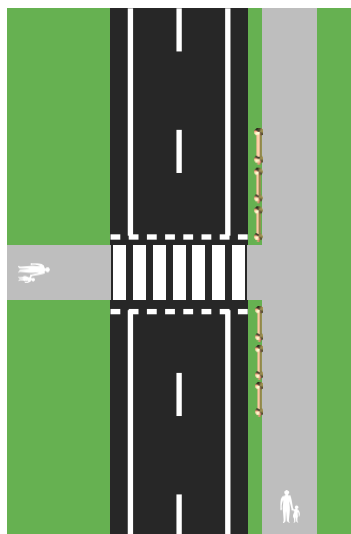
	0m to <1m Wide	
Road edge is 0m to <1.0m wide. Rule of thumb – A narrow paved shoulder should be recorded wherever an edge marking is present even if the edge line is very close to the pavement edge.		
	1m to <2.4m Wide	
Road edge is 1m to less than 2.4 m wide.		
	≥2.4m Wide	
Road edge is ≥ 2.4m wide. Rule of thumb: A wide shoulder is wider than a car.		



7.3 Pedestrian channelization

Record if there is fencing or other feature/s present (e.g. garden beds) that direct pedestrians toward crossing facilities on one or both sides of the road.

Coding options

	Not Present	
	No pedestrian channelization present.	
	Present	
	Pedestrian channelization present.	



8 Pedestrian crossing

8.1 Crossing on main and side road

Record if there is a pedestrian crossing facility on the inspected road or a side road.

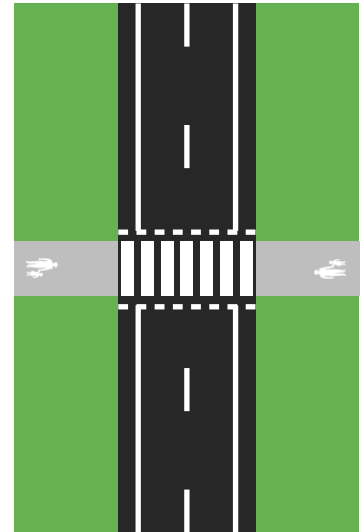
If a crossing is present, further information about it will be requested.

Coding options



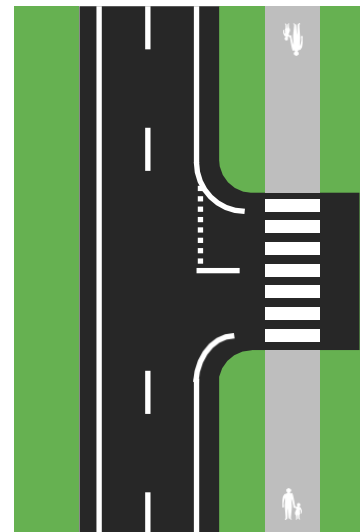
Present on main road

A pedestrian crossing facility is present on the main road.

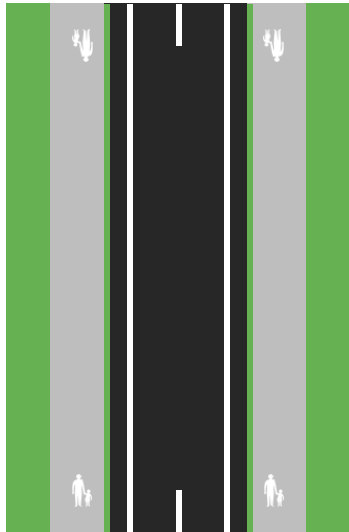


Present on side road

A pedestrian crossing facility is present on a side road.





	Not present	
No pedestrian crossing facility on either a main road or a side road.		



9 Pedestrian crossing type

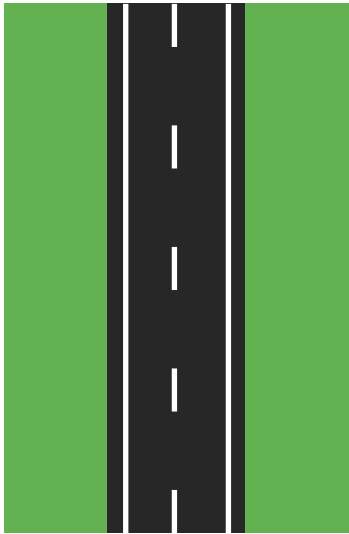
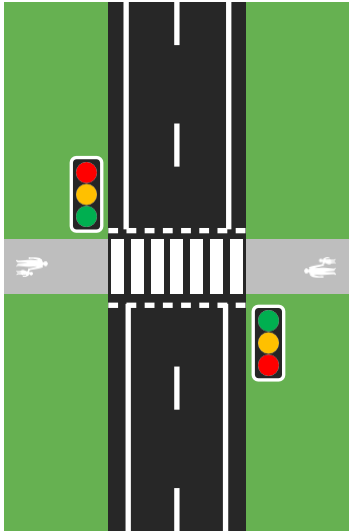
9.1 Crossing features

Record details for the pedestrian crossing recorded.

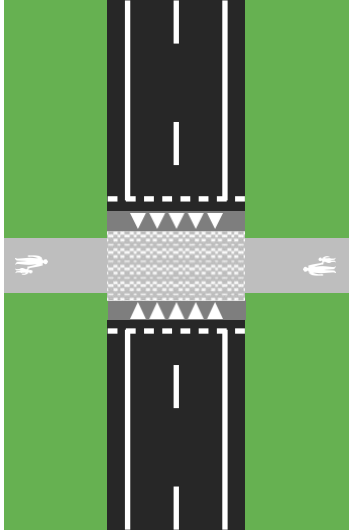
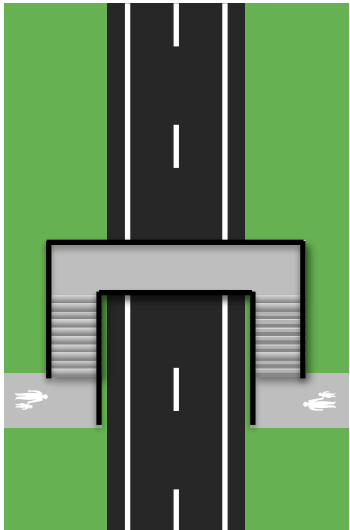
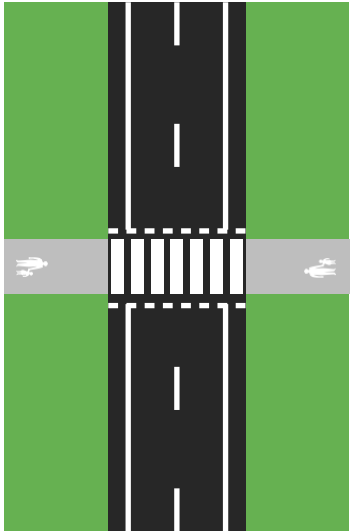
Notes:

The selected crossing feature must reflect the infrastructure available on the assessed road. If there is no crossing facility, the option *None* must be selected.

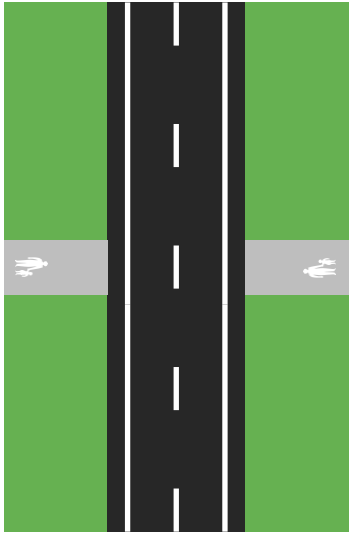
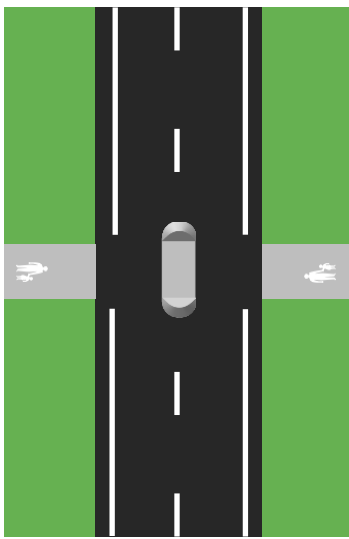
Coding options

None No crossing facility is offered for pedestrians.	
Lights The crossing has traffic lights to control pedestrians and vehicles movement.	



	Raised	
	The crossing is raised from the road surface, meaning that vehicles have to slow down at the crossing.	
	Bridge or Tunnel	
	The crossing is higher or lower than the road and hence does not mix pedestrians and vehicles.	
	Marked	
	The crossing has markings on the road.	



	Unmarked	
	The crossing has no markings on the road.	
	Refuge	
	A central refuge island is provided for pedestrians. This may be a purpose-built island or refuge provided as part of a physical median.	



9.2 Crossing quality

Record how well the pedestrian crossing can be seen by drivers, or if there are warning signs present.

Notes:

The three primary factors to be considered in coding quality of crossing are:

- Signing
- Markings
- Good visibility

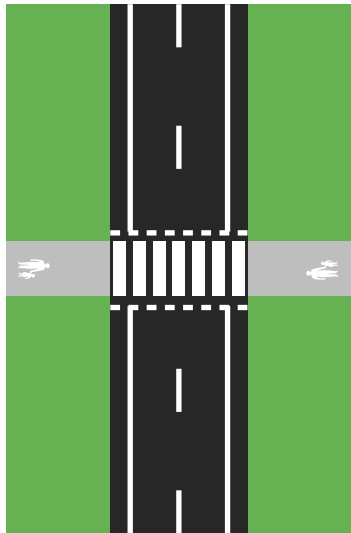
An effective pedestrian crossing is one where:

- Vehicle drivers are required to stop and give pedestrian right-of-way (and do so)
- The facility is clearly visible and can be anticipated by vehicle drivers, and
- The facility is not obstructed by parked vehicles, street furniture or other items.

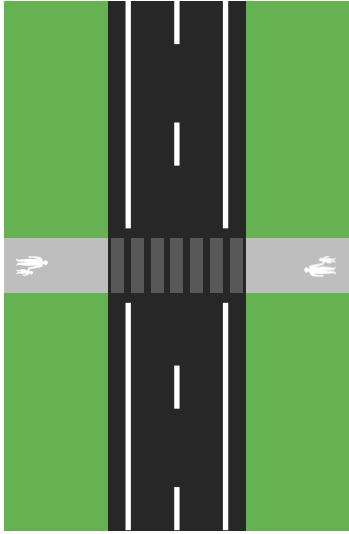
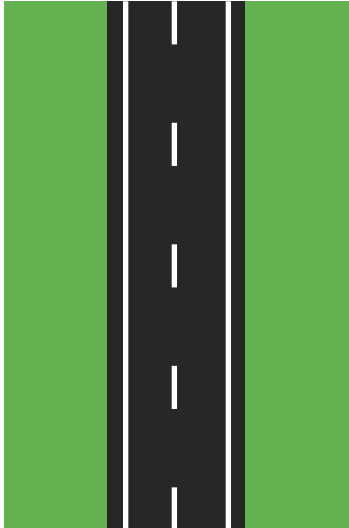
A pedestrian crossing is ineffective where for example:

- Drivers may brake suddenly or fail to see the facility due to the road's curvature, speed, downward gradient, poor sight distance or other factor,
- Pedestrians are entering the facility between parked cars and cannot be seen by approaching drivers or
- Green pedestrian signals do not provide right-of-way to pedestrians.

Coding options

	Adequate	
	Signing, marking & visibility enable a driver to be aware in good time of the presence of a pedestrian crossing.	



	Poor	
	Crossing facility is ineffective. Rapid or unexpected speed adjustments might be needed to avoid risk to pedestrians.	
	No crossing	
	There is no pedestrian crossing facility in the assessed location.	



10 Flow

10.1 Vehicles per day

Record the total number of vehicles that would use the road in a day.

10.2 Pedestrian crossing flow per hour

Record the average number of pedestrians that:

- Cross the road during the busiest hour of the day
- Walk along the left side of the road during the busiest hour of the day, and
- Walk along the right side of the road during the busiest hour of the day.

Coding options

	0
	1 to 5
	6 to 25
	26 to 50
	51 to 100
	101 to 200
	201 to 300
	301 to 400
	401 to 500
	501 to 900
	900+



10.3 Motorcycle percent (%)

Record the percentage of motorcycles as a proportion of total vehicles per day.

Coding options

	0%
	1-5%
	6-10%
	11-20%
	21-40%
	41-60%
	61-80%
	81-99%
	100%

10.4 Heavy vehicle percent (%)

Record the percentage of heavy vehicles as a proportion of total vehicles per day.

Coding options

	0-5%
	6-10%
	11-15%
	16-20%
	21-30%
	31-40%
	>40%



11 Intersections

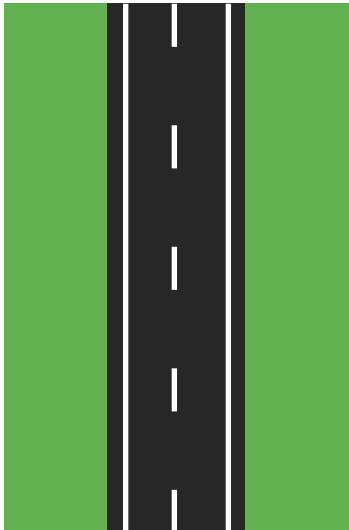
11.1 Intersection present

Record if there is an intersection at the location.

Notes

If an intersection is recorded, the system will also ask about its features.

Coding options

	Intersection	
	There is an intersection at the location. It can be any type of intersection, not just a 3-leg intersection as shown on the icon.	
	No Intersection	
	There is no intersection at the location.	



12 Intersection details

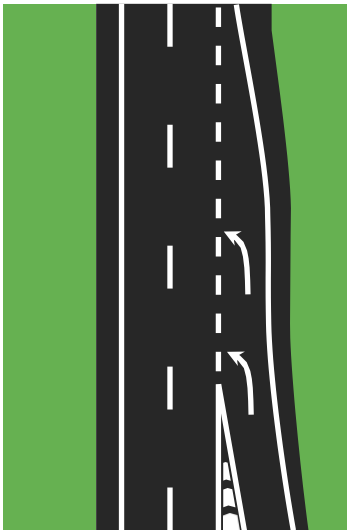
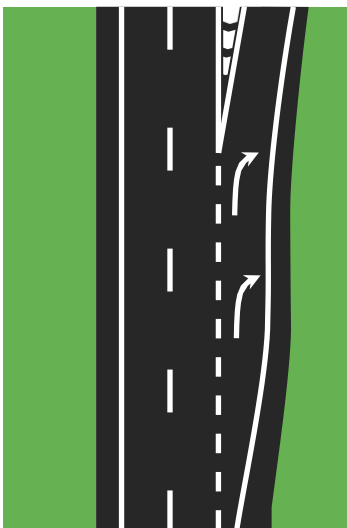
12.1 Intersection type

Record key features of the intersection.

Notes

If a 3-leg or 4-leg intersection is coded, further options are required. In the website, the option that combines the type of intersection and all the additional features present must be selected.

Coding options

	Merge lane	
	Traffic joins the road being assessed via a merge lane. This is typically the case at split level intersections.	
	Diverge lane	
	Traffic exits the road being assessed via diverge lane. This is typically the case at split level intersections.	
	3 legs	



The intersection has 3 legs.



4+ legs

The intersection has 4 or more legs.

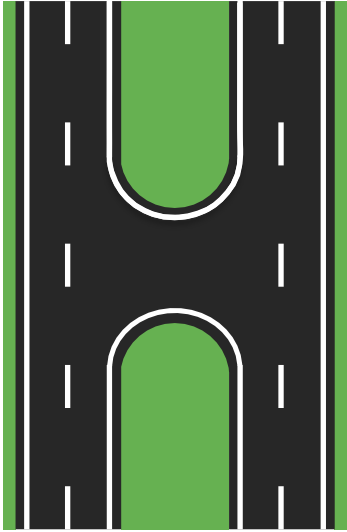
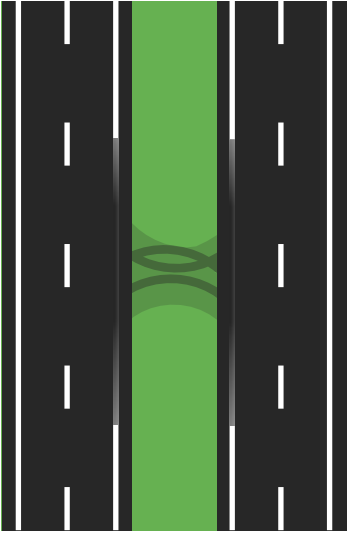


Roundabout

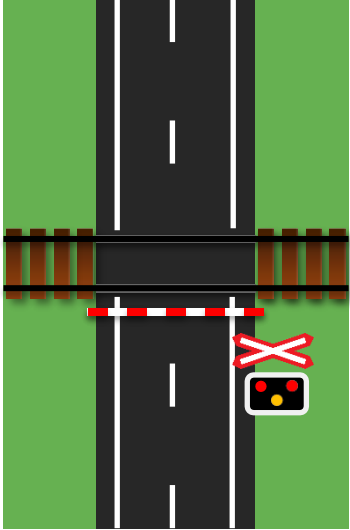
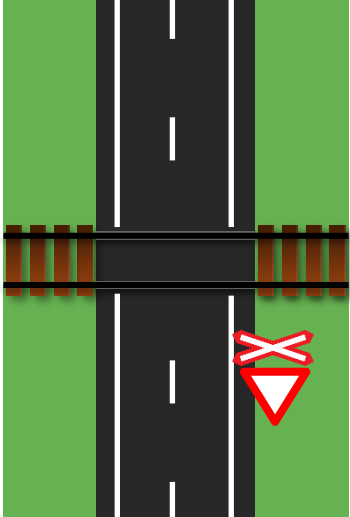
Circular intersection with flow in a single direction around a central island.





	Mini roundabout	
	Small roundabout – typically found in low-speed urban areas.	
	Formal U-turn	
	A formal (designated) median crossing that allows vehicles to cross over the median.	
	Informal U-turn	
	A median crossing that has developed informally overtime through use.	



	Railway crossing with active signals	
	An at-grade railway crossing with a physical device to warn of an approaching train (flashing lights and/or boom gates).	
	Railway crossing with passive signage only	
	An at-grade railway crossing with give way or stop signage only (or no signage).	



12.2 Turn lane

Record whether a 3 or 4 leg intersection has a turn lane.

Notes

Only record turn lanes for 3 leg or 4 leg intersections. In the website, the option that combines the type of intersection with or without turn lanes present must be selected.

Coding options

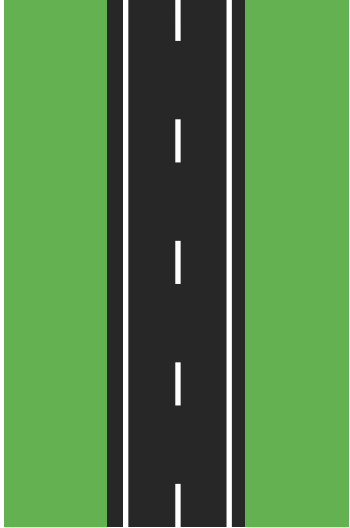
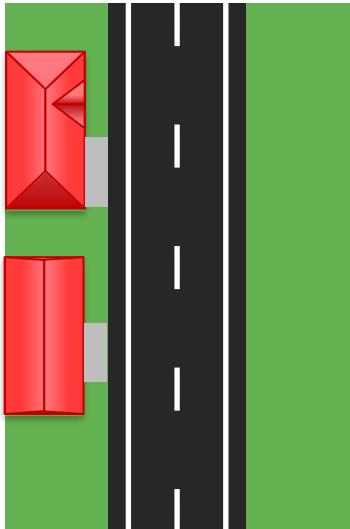
	Turn Lane Present	
	A turn lane that allows vehicles to wait in the middle of the intersection before turning across the vehicles travelling in the opposite direction.	



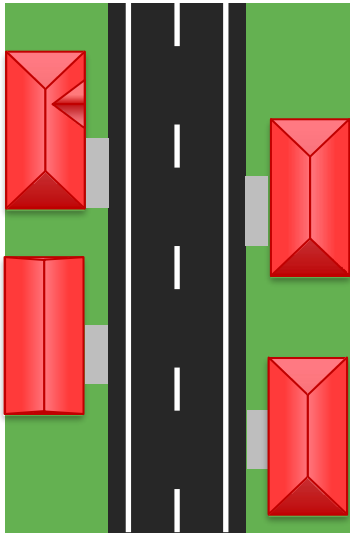
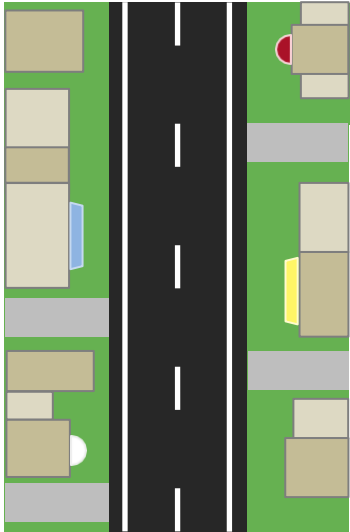
12.3 Driveways

Record the number of driveways and entrances where vehicles can enter or exit the roadway. This includes commercial and residential driveways and minor access lanes.

Coding options

	Not applicable	
	No residential or commercial access points.	
	1 or 2 Driveways	
	Fewer than 3 residential access points or an access point that serves three or fewer residential properties. May also include minor farm accesses and other minor tracks used by motor vehicles to gain occasional access to the main carriageway.	



	>2 Driveways	
	Three or more residential access points or an access point that serves three or more residential properties.	
	Commercial Access	
	Commercial access point present. Examples include service stations, shops and roadside cafes, and rest stops.	

12.4 Intersection side road flow

Record the total number of vehicles that would use the intersecting road in a day.



12.5 Intersection quality

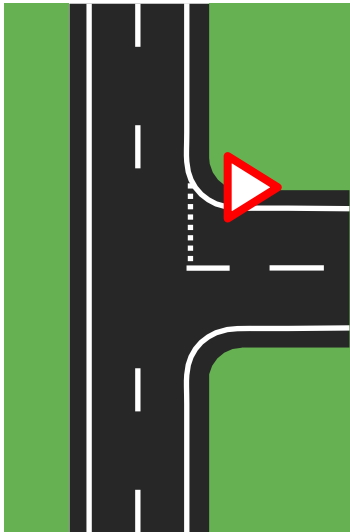
Record if the intersection has appropriate design, advance warning, signing, and markings.

Notes:

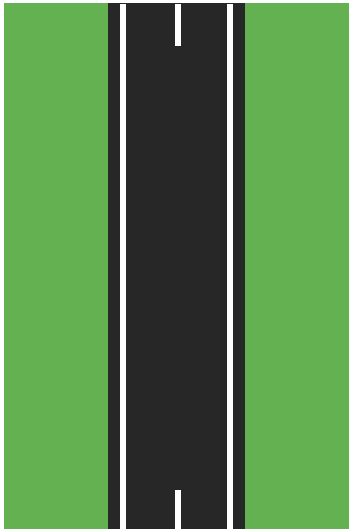
Factors resulting in a 'poor' intersection may include:

- Lack of advance signage and marking on an intersection approach where the intersection is not clearly visible to approaching drivers (such as where approach visibility is limited).
- Very short merge lanes.
- Intersection geometry not designed in accordance with posted speed limit.

Coding options

	Adequate	
Intersection design is adequate and necessary signage and markings are present.		
	Poor	
Intersection design is poor, or necessary signage and markings are absent or obstructed with limited visibility, or where the angle with the intersecting leg is small enough to reduce the visibility.		



NA	No intersection	
There is no intersection in the assessed location.		



13 Curve

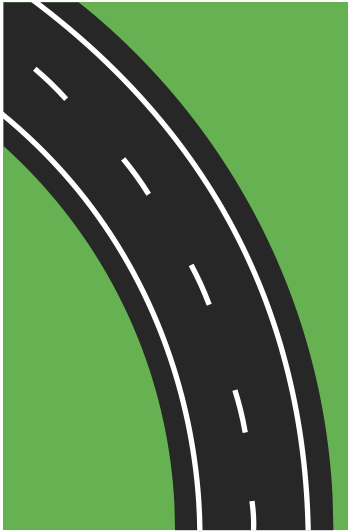
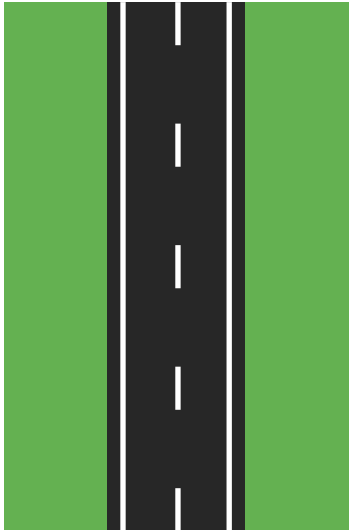
13.1 Curve

Record if there is a curve at the location.

Notes

In the website, the option that combines the presence and type of curve must be selected.

Coding options

	Curve	
There is a curve at the location.		
	No Curve	
There is no curve at the location.		

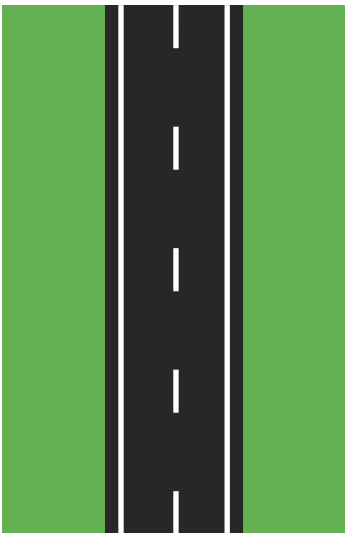


14 Curve details

14.1 Curve type

Record details about the curve.

Coding options

	Straight	
The road section contains no curve or long curves.		

	Moderate	
The road has a fairly tight curve.		



	Sharp	
The road has sharp curves, nearly 45°.		

	Very sharp	
The road has curves between 45° and 90° which can only be driven at low speeds.		

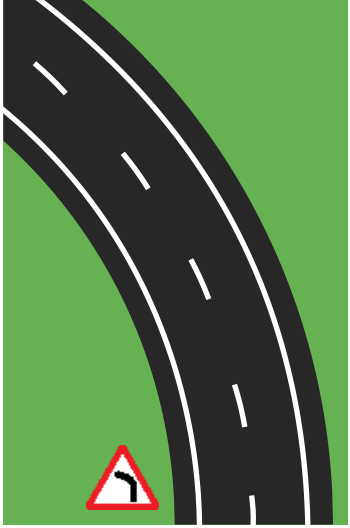


14.2 Curve quality

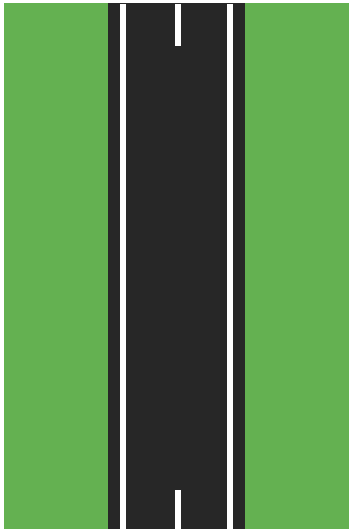
Record how easy it is for a driver to judge how sharp a curve is and whether it can be driven safely.

The quality of the curve will reflect the extent to which signage and markings help the driver to judge the correct curvature, and the visibility in advance of, and around, the curve.

Coding options

☒	Adequate	
	Signage, markings and visibility where needed, enable a driver to judge curvature.	
☐	Poor	
	Rapid or unexpected speed adjustments are needed to negotiate the curve. Lack of advanced signage. Missing or poorly maintained road markings.	



NA	No curve	
There is no curve in the assessed location.		



15 Speed

15.1 Speed limit

Record the posted speed limit for vehicles on the road.

Do not attempt to record the speed of the traffic using the road.

Do not record temporary speed limits at roadworks or advisory speeds.



Coding options

	≥150	Speed limit is 150km/h / 150mph or greater
	140	Speed limit is 140km/h / 140mph
	130	Speed limit is 130km/h / 130mph
	120	Speed limit is 120km/h / 120mph
	110	Speed limit is 110km/h / 110mph
	100	Speed limit is 100km/h / 100mph
	90	Speed limit is 90km/h / 90mph
	80	Speed limit is 80km/h / 80mph
	70	Speed limit is 70km/h / 70mph
	60	Speed limit is 60km/h / 60mph
	50	Speed limit is 50km/h / 50mph
	40	Speed limit is 40km/h / 40mph
	30	Speed limit is 30km/h / 30mph or less



15.2 Speed limit unit

Record what units are used (kilometres or miles per hour) for the speed limits recorded.

15.3 Operating speed

In the web application, record the operating speed registered at that location.

Notes:

The unit of the operating speed will change based on the speed limit unit selected.

Coding options (km/h)

-10 km/h	-10 km/h	Most vehicles are travelling 10km/h below the speed limit
-5 km/h	-5 km/h	Most vehicles are travelling 5km/h below the speed limit
0 km/h	0 km/h	Most vehicles are travelling at the speed limit
+5 km/h	+5 km/h	Most vehicles are travelling 5km/h above the speed limit
+10 km/h	+10 km/h	Most vehicles are travelling 10km/h above the speed limit
+15 km/h	+15 km/h	Most vehicles are travelling 15km/h above the speed limit



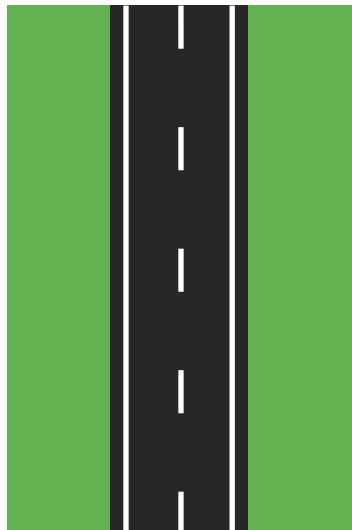
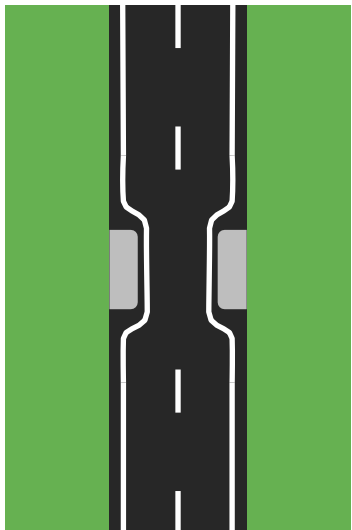
15.4 Speed management

Record the presence of road features that will typically reduce the operating speed by 5-10km/h below the speed limit.

Notes:

Traffic calming features include kerb buildouts, speed bumps, raised tables, entry treatments, speed cushions, chicanes and modified intersections.

Coding options

	Not present	
No traffic calming or speed management features present.		
	Present	
Traffic calming/speed management features are present to a degree sufficient to reduce the operating speed by 5-10km/h below the speed limit. For example: • Kerb build outs • Speed table • Speed bump		



Version

Date	Version	Update
Sept 2017	1.0	Document created.
Jan 2019	1.1	Added 'None' category for property access point.
May 2019	1.2	Sequence of attributes reorganised.
Mar 2020	1.3	Updates related to SR4S Global Launch.
May 2020	1.4	Minor updates.
Oct 2020	1.5	New description of crossing quality and right and left side of the road.
June 2021	1.6	General edits and minor changes to attribute and sub-attribute labels. Land use 'residential' definition updated. Pedestrian fence attribute is locked for review. Diagrams updated for grade, median, school zone, school crossing supervisor and road edge.
June 2023	1.7	Broad updates to reflect changes to the pedestrian model. New categories include 'Diverge lane' in intersection type; 'Shared path' in sidewalks, and 'Unmarked crossing' in pedestrian crossing features. Intersection channelization has been removed (has no impact on the pedestrian model). 'Pedestrian channelisation' attribute replaces 'Pedestrian fencing'. Coding guidance changes required for intersection side road flow and for crossings with pedestrian refuges which are integrated into the road median. Motorcycle and heavy vehicle percentage have been introduced into Flow. Signals have been removed from intersection details (has no impact on the pedestrian model). Diagrams simplified and updated.